

Retrieval-Augmented Generation: An Architectural Guide

1. What is Retrieval-Augmented Generation?

Retrieval-Augmented Generation (RAG) is an architecture that combines a retrieval system with a generative language model. Instead of relying solely on the knowledge encoded in the model's parameters, RAG retrieves relevant external documents at query time and supplies them as context to the language model. This significantly reduces hallucinations and allows the system to answer questions about private, domain-specific, or recently updated information.

2. The Two-Stage Pipeline

A RAG system generally consists of two stages: an indexing stage and a query stage. During indexing, documents are collected, split into smaller chunks, converted into vector embeddings, and stored in a vector database such as FAISS. During the query stage, the user's question is embedded using the same model, and the vector database is searched for the most semantically similar chunks, which are then passed to a language model along with the question to produce a grounded answer.

3. Chunking Strategy

Document chunking is critical to retrieval quality. If chunks are too large, the retrieved context may contain irrelevant information that dilutes the language model's focus. If chunks are too small, important context may be split across multiple chunks and lost. A common approach is to use a recursive character text splitter with a chunk size of around 1000 characters and an overlap of 100 to 200 characters, which helps preserve context across chunk boundaries.

4. Dual-Model Architectures

Some advanced RAG systems use two language models instead of one. A smaller or faster model is used first to summarize and condense the retrieved chunks along with any conversation history, reducing the total token count. A second, more capable model then receives this condensed context and generates the final, detailed answer, often including citations back to the original source documents. This approach can reduce cost and latency while maintaining answer quality.

5. Evaluating RAG Systems

Common evaluation metrics for RAG systems include retrieval precision and recall (whether the correct chunks were retrieved), answer faithfulness (whether the generated answer is actually supported by the retrieved context), and answer relevance (whether the answer actually addresses the user's question). Tools such as RAGAS are often used to automate this evaluation process using LLM-based judges.

6. Conclusion

Retrieval-Augmented Generation has become a foundational technique for building reliable, up-to-date AI assistants that can reason over private or rapidly changing document collections, making it a core architecture in modern generative AI applications.